

Solution to Exercise 05 — Analysis on Manifolds, 80416

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Question 1

Consider the n sphere with radius 1 in $\mathbb{R}^{n+1}$:

$$S^n := \{x \in \mathbb{R}^{n+1} \mid \|x\| = 1\}$$

Denote $N = e_{n+1}$ where $(e_1, \dots, e_{n+1})$ is the standard basis for $\mathbb{R}^{n+1}$. Define $\pi : S^n \setminus \{N\} \rightarrow \mathbb{R}^n$ in the following way: for every $p \in S^n \setminus \{N\}$ we draw the straight line from N to p and we define $\pi(p)$ to be the intersection of that line with the space

$$\{x \in \mathbb{R}^{n+1} \mid x_{n+1} = 0\}$$

Subquestion 1

We will give an explicit formula for π and we will show that it is continuous (we call π the Stereographic Coordinates for S^n).

Proof: By definition, the north pole is $N = (0, 0, \dots, 0, 1)$.

The straight line L passing through N and p can be parametrized by a scalar $t \in \mathbb{R}$:

$$L(t) = N + t(p - N) = (0, \dots, 0, 1) + t(x_1, \dots, x_n, x_{n+1} - 1)$$

Hence

$$L(t) = (tx_1, tx_2, \dots, tx_n, 1 + t(x_{n+1} - 1))$$

From the definition, $\pi(p)$ is the intersection with the last coordinate of $L(t)$ to zero so we have

$$1 + t(x_{n+1} - 1) = 0 \implies t(1 - x_{n+1}) = 1 \implies t = \frac{1}{1 - x_{n+1}}$$

We first notice that the above is well defined since $x_{n+1} \neq 1$ from the definition of S^n .

Substituting back into the first n coordinates will give us $\pi : S^n \setminus \{N\} \rightarrow \mathbb{R}^n$ by

$$\pi(p) = \left(\frac{x_1}{1 - x_{n+1}}, \frac{x_2}{1 - x_{n+1}}, \dots, \frac{x_n}{1 - x_{n+1}} \right)$$

We notice that each component is a rational function that never vanish and is continuous as a quotient of continuous functions. In $\mathbb{R}^n$ a map is continuous if and only if each of its coordinate is continuous hence π continuous. □

Subquestion 2

We will prove that π is bijective and find an explicit formula for $\pi^{-1} : \mathbb{R}^n \rightarrow S^n \setminus N$.

Proof: If we construct its inverse map we are done.

Let $y = (y_1, \dots, y_n) \in \mathbb{R}^n$, we seek a unique point $p = (x_1, \dots, x_{n+1}) \in S^n \setminus \{N\}$ such that $\pi(p) = y$ which means for $i = 1, \dots, n$

$$y_i = \frac{x_i}{1 - x_{n+1}} \iff x_i = y_i(1 - x_{n+1})$$

Since $p \in S^n$, its coordinates must satisfy the equation of the sphere:

$$\sum_{i=1}^n x_i^2 + x_{n+1}^2 = 1$$

So in our case

$$\sum_{i=1}^n (y_i(1 - x_{n+1}))^2 + x_{n+1}^2 = 1 \iff (1 - x_{n+1})^2 \sum_{i=1}^n y_i^2 + x_{n+1}^2 = 1 \iff \sum_{i=1}^n y_i^2 = \|y\|^2 \iff \|y\|^2(1 - x_{n+1})^2 + x_{n+1}^2 = 1$$

So

$$\|y\|^2(1 - 2x_{n+1} + x_{n+1}^2) + x_{n+1}^2 - 1 = 0 \iff (\|y\|^2 + 1)x_{n+1}^2 - 2\|y\|^2x_{n+1} + (\|y\|^2 - 1) = 0$$

We know that $x_{n+1} = 1$ is a root of this equation but our domain is $S^n \setminus \{N\}$ hence $x_{n+1} \neq 1$.

We use Vieta's formulas from Algebraic Structures 2 and gets that the other valid root must be

$$x_{n+1} = \frac{\|y\|^2 - 1}{\|y\|^2 + 1}$$

Hence

$$1 - x_{n+1} = 1 - \frac{\|y\|^2 - 1}{\|y\|^2 + 1} = \frac{\|y\|^2 + 1 - \|y\|^2 + 1}{\|y\|^2 + 1} = \frac{2}{\|y\|^2 + 1}$$

Substituting this back into our expression for x_i :

$$x_i = y_i(1 - x_{n+1}) = \frac{2y_i}{\|y\|^2 + 1}$$

Thus, we have found a unique, explicitly defined point p for every $y \in \mathbb{R}^n$. This gives us our inverse function $\pi^{-1} : \mathbb{R}^n \rightarrow S^n \setminus \{N\}$ by

$$\pi^{-1}(y) = \left(\frac{2y_1}{\|y\|^2 + 1}, \dots, \frac{2y_n}{\|y\|^2 + 1}, \frac{\|y\|^2 - 1}{\|y\|^2 + 1} \right)$$

□

Subquestion 3

We will prove using the Stereographic Coordinates that we can cover the sphere using two local parametrizations $\pi_i : \mathbb{R}^n \rightarrow S^n$ for $i \in \{1, 2\}$.

Proof: We have already established the inverse stereographic projection from the North pole $N = (0, \dots, 0, 1)$, which gives us our first local parametrization:

$$\pi_1 = \pi^{-1} : \mathbb{R}^n \rightarrow S^n \setminus \{N\}.$$

This parametrization covers the entire sphere except for the North pole N .

To cover N , we define a second stereographic projection, this time from the South pole $S = (0, \dots, 0, -1)$.

By symmetry (or by repeating the exact derivation with S instead of N), the inverse of the stereographic projection from the South pole provides a second local parametrization:

$$\pi_2 : \mathbb{R}^n \rightarrow S^n \setminus \{S\}.$$

The image of π_1 is $S^n \setminus \{N\}$, and the image of π_2 is $S^n \setminus \{S\}$.

Since $N \neq S$, the union of their images is:

$$(S^n \setminus \{N\}) \cup (S^n \setminus \{S\}) = S^n$$

Thus, S^n is entirely covered by the two local parametrizations π_1 and π_2 .

□

Subquestion 4

We will prove that the sphere could not be covered with one local parametrization $r : U \rightarrow S^n$ where $U \subseteq \mathbb{R}^n$ is open.

Proof: By contradiction, assume that there exists a single local parametrization $r : U \rightarrow S^n$ that covers the entire sphere, where $U \subseteq \mathbb{R}^n$ is an open set.

By the definition of a local parametrization, r must be a homeomorphism from U onto its image, which by our assumption is S^n so this implies that U and S^n are homeomorphic.

From its definition, S^n is a closed and bounded subset hence by Heine-Borel theorem its compact.

Since $r^{-1} : S^n \rightarrow U$ is a homeomorphism and S^n is compact, its image U must also be a compact subset of $\mathbb{R}^n$.

Since any compact subset of $\mathbb{R}^n$ is closed and bounded we get that U is both open and closed in $\mathbb{R}^n$.

We saw that $\mathbb{R}^n$ is a connected topological space and the only subsets that are both open and closed in a connected topological space are the $\emptyset$ and $\mathbb{R}^n$ itself and since S^n is not empty, U cannot be empty, so we must have $U = \mathbb{R}^n$.

But $\mathbb{R}^n$ is unbounded which means that it is not compact and that is a contradiction.

□

Question 2

Question 3

Question 4

Subquestion 1

We will prove that $C = \{(x, y, a\sqrt{x^2 + y^2}) \in \mathbb{R}^3 \mid x, y \in \mathbb{R}, a > 0\}$ is not a manifold of dimension 2 in $\mathbb{R}^3$.

Proof: If it were a smooth manifold, it would have to be locally representable as a smooth graph over at least one of the three coordinate planes (xy , xz , or yz) so we look at the origin $p = (0, 0, 0)$.

Assume for contradiction that C is a smooth manifold of dimension 2.

By the equivalent graph definition of a manifold, there must exist an open neighborhood of p in C that can be represented as the graph of a smooth function over an open set in one of the coordinate planes (xy , xz , or yz).

1. If it is a graph over the xy plane

By the definition of the set, C is globally the graph of the function $f(x, y) = a\sqrt{x^2 + y^2}$ so f must be smooth (and thus differentiable) at $(0, 0)$, but

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{t \rightarrow 0} \frac{f(t, 0) - f(0, 0)}{t} = \lim_{t \rightarrow 0} \frac{a\sqrt{t^2}}{t} = \lim_{t \rightarrow 0} \frac{a|t|}{t}$$

Since the limit evaluates to a from the right and $-a$ from the left, the partial derivative does not exist. So f is not smooth at the origin.

2. If it is a graph over the xz plane or yz plane: We show for the plane xz and for the plane yz it is from the same reasoning.

Suppose C can be represented as $y = g(x, z)$ for some smooth function $g : U \rightarrow \mathbb{R}$, where $U \subseteq \mathbb{R}^2$ is an open neighborhood of $(0, 0)$ in the xz -plane.

By definition, every point in C has a z -coordinate $z = a\sqrt{x^2 + y^2} \geq 0$.

This means the projection of C onto the xz -plane is restricted to the upper half-plane $z \geq 0$.

Since U must be an open set containing $(0, 0)$, it must contain points with $z < 0$. However, there are no points in C that map to negative z values, meaning C cannot be written as a graph over an open set U around $(0, 0)$.

Since C cannot be locally represented as a smooth graph over any the coordinate planes at the origin it is a contradiction to the assumption and C is not a manifold of dimension 2 in $\mathbb{R}^3$. $\square$